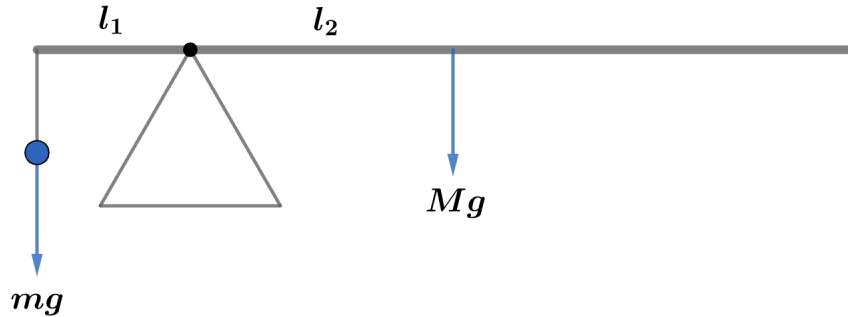


2007 F=ma Exam: Problem 12

Kevin S. Huang



Balancing torques around the fulcrum, we have

$$mgl_1 = Mgl_2$$

using the fact that we can put all the weight of the measuring stick at its center of mass to find the equivalent torque. Since $l_1 = 0.2\text{ m}$, we have $l_2 = 0.5\text{ m} - l_1 = 0.3\text{ m}$. Thus,

$$M = \frac{l_1}{l_2}m = \frac{0.2\text{ m}}{0.3\text{ m}}(2\text{ kg}) = 1.33\text{ kg}$$

so the answer is .